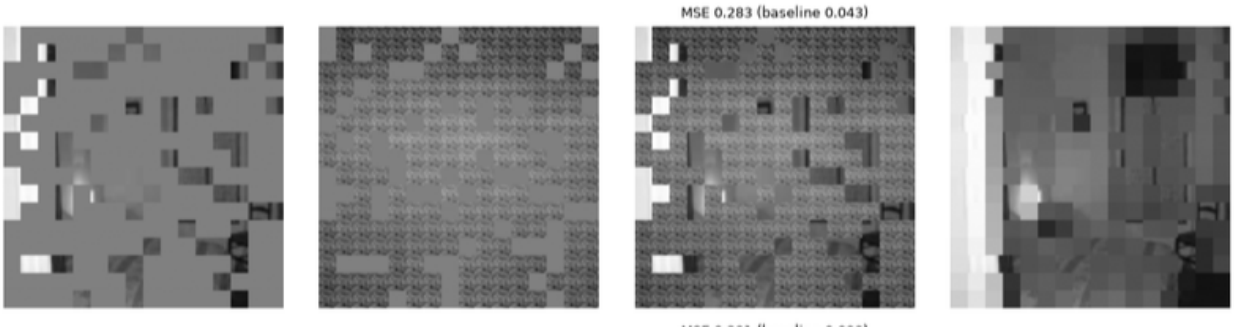
Assignment 2: Self-Supervised Learning

For Assignment 2, I explored the applicability of the Masked Autoencoder (MAE), introduced by He et al. (2022), to our dataset. The MAE architecture consists of an encoder and a decoder: the encoder sees only a random subset of image patches, and the decoder reconstructs the masked patches from the encoder's output. All of MAE's training was unlabeled. MAE served as the pretraining stage for our model. For classification, the pretrained encoder was fine-tuned with a linear classification layer on top. The dataset contains 16 evenly distributed classes totaling 2,800 images.

I chose MAE because it is prevalent in much larger models and architectures, and I wanted to understand its design and investigate how it behaves on a small dataset and limited hardware. All experiments ran on a Linux home server with a 6 GB NVIDIA GTX 1660 Ti, and hardware was a severe limiting factor. Patch size was limited to a minimum of 6, resolution was capped at 150×150 pixels, and the encoder was limited to 6 layers, 8 heads, a 128-dimensional embedding, and a 512-dimensional MLP. Runs at the highest possible settings averaged one hour and thirty minutes. The resulting model reached 80% training accuracy and 63% validation accuracy, which indicates overfitting.

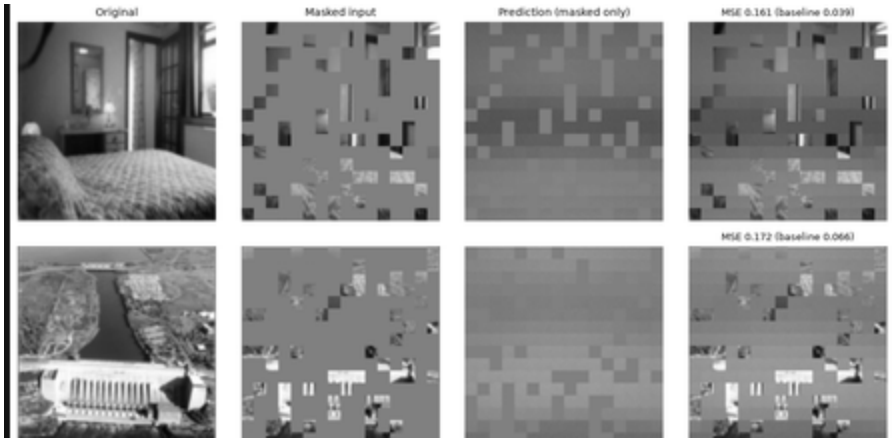
Subsequent experiments focused on architectural improvements and parameter tuning. MAE is designed for training at scale, so it suffers on a smaller dataset. Experiments were limited to a maximum of 100 epochs, with MAE pretraining closer to 75. I separated the iterations by the specific problem each one addressed. MAE reconstructed lighting and brightness well (Figures 1 and 2), but early training produced noisy reconstructions of the missing patches.

(Figure 1)



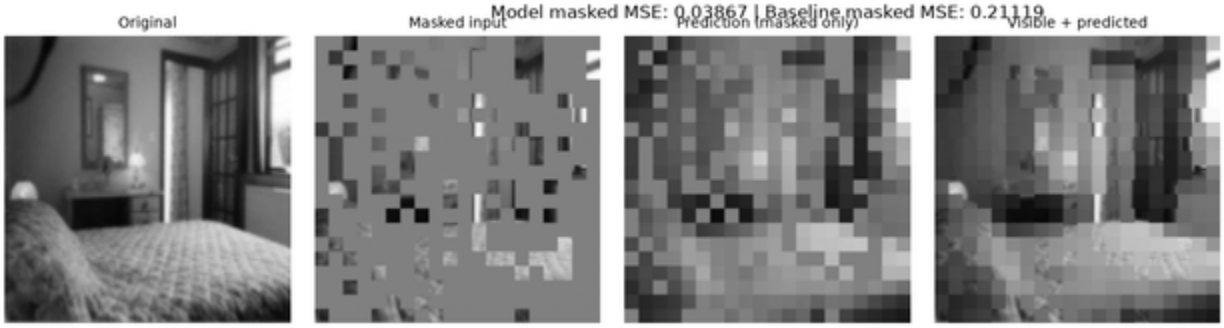
The next iteration focused on removing that noise, which was caused by an undertrained decoder. At this stage the decoder mirrored the encoder's dimensions. The improved predictions are shown in Figure 2.

(Figure 2)



The following improvements targeted the MAE loss function. I used per-patch normalized targets (norm_pix): each target patch is normalized by its own mean and variance, which prevents the model from lowering the loss by predicting overall brightness. This forced the model to learn richer structure in the images, such as edges (Figure 3).

(Figure 3)



Through tuning and augmentation, I increased the mask ratio from 75% to 80% and the MAE pretraining epochs to 80. I also implemented CutMix through a DataLoader collate_fn, which applies a custom function to each whole batch. Training epochs were increased to 75, to accommodate for Cut-Mix.

My biggest limitations were the hardware and dataset. MAE provided a clear benefit, visible during warmup: validation accuracy jumped to about 49%, close to the 51% the basic transformer reached only after full training. One limitation was, the dataset being in grayscale, MAE performs better using RGB. My implementation also utilized a larger decoder, whereas the original paper implemented a smaller decoder. In conclusion, MAE pretraining was a beneficial step for this dataset. Fine-tuning from the pretrained encoder reached about 49% validation accuracy within the warmup phase, roughly matching the baseline transformer's final accuracy of 51%, and the final model reached 70% test accuracy. The reconstruction figures show the pretraining progressing from noisy, brightness-only predictions toward structure such as edges, suggesting the encoder learned more than low-level intensity. These gains came with limits: the small dataset and limited hardware kept the model small, and overfitting remained the main obstacle.

Sources:

K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, “Masked autoencoders are scalable vision learners,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 16000–16009.

Github Link